G. NARAYANAMMA INSTITUTE OF TECHNOLOGY AND SCIENCE

(AUTONOMOUS) (For Women)

Shaikpet, Hyderabad - 500 104

Department of Computer Science and Engineering

IV B.Tech. I Semester Academic Year 2026-27

Assignment-2

SUBJECT: Data Science Using R

SUBJECT CODE: 127EY **Max Marks: 5**

BRANCH: CSE **Date: 6-10-2026**

Blooms Taxonomy Levels:
Level 1 – Remembering, Level 2 – Understanding, Level 3 – Applying, Level 4 – Analyzing, Level 5 – Evaluating, Level 6 – Creating

Answer all questions.

<table>
<tr><th>S. No.</th><th>Question</th><th>CO</th><th>BT Level</th></tr>
<tr><td>1</td><td>a. Differentiate between memorization-based and generalization-based (non-memorization) methods in machine learning. Provide suitable examples for each.
b. What is the significance of KDD in data mining and Machine Learning with reference to knowledge discovery from data.</td><td>CO2
CO2</td><td>L1
L2</td></tr>
<tr><td>2</td><td>a. Describe the concept of distance measures in cluster analysis and their role in grouping data points. Compute two clusters (by taking two data points) for two iterations for the given data using K means algorithm
<table>
<tr><th>Record</th><th>A</th><th>B</th></tr>
<tr><td>R1</td><td>1</td><td>1</td></tr>
<tr><td>R2</td><td>1.5</td><td>2</td></tr>
<tr><td>R3</td><td>3</td><td>4</td></tr>
<tr><td>R4</td><td>5</td><td>7</td></tr>
<tr><td>R5</td><td>3.5</td><td>5</td></tr>
<tr><td>R6</td><td>5</td><td>5</td></tr>
<tr><td>R7</td><td>3.5</td><td>4.5</td></tr>
</table>
b. Explain hierarchical clustering and perform hierarchical clustering (bottom up) for the above data up to two iterations using complete link coverage.</td><td>CO6</td><td>L2</td></tr>
<tr><td>3</td><td>a. Explain how the process of building a model using a single predictor differs from building a model using multiple predictors. Demonstrate with an example</td><td>CO6
CO4</td><td>L2
L5</td></tr>
</table>

	b. What is the Receiver Operating Characteristic (ROC) curve, and how is it used to evaluate the performance of a logistic regression model?		
4	Compare and contrast simple linear regression and multiple linear regression, Generalized Linear Model (GLM) and when is each appropriate to use?	CO3	L4
5	a. What does ARIMA stand for, and what are the components of an ARIMA model? outline some common applications of time series analysis in real-world scenarios? b. Write a short note on decomposing time series data.	CO5 CO5	L3 L4